

JEE MAIN REVISION MASTER: HYDROGEN AND IT'S COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Hydrogen — Core Principles & Governing Laws	9 PYQs	Core Principle
Standard	Hydrogen — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics
Standard	Hydrogen — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hydrogen — Core Principles & Governing Laws	$30\%(w/v)H_2O_2$, $\text{CO}(g)+H_2(g)$, Ca^{2+}
Hydrogen — Applied Cases, Reactions & Traps	$C(s)+H_2O(g) \rightarrow CO(g)+H_2(g)$, $(CO+H_2)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hydrogen — Core Principles & Governing Laws (High Yield — 9 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Hydrogen.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Hydrogen.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Hydrogen PYQs.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following properties is generally not applicable to ionic hydrides?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The correct decreasing order of boiling point of hydrogen halides is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The composition of water gas is..." [Ans: N/A]

3.2 Hydrogen — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Hydrogen.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2017 Shift] Q#1 (MCQ): "In the manufacture of hydrogen from water gas ($\text{CO} + \text{H}_2$), which of the following is correct statement?..." [Ans: N/A]

3.3 Hydrogen — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Hydrogen.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following properties is generally not applicable to ionic hydrides?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "The correct decreasing order of boiling point of hydrogen halides is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

The correct decreasing order of boiling point of hydrogen halides is

- (A) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (B) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (C) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
- (D) $\text{HI} > \text{HF} > \text{HBr} > \text{HCl}$

Correct Answer: None

Solution Analysis: HF is hydrogen bonded, thus it has highest boiling point. The remaining hydrogen halides are gaseous and their boiling points depend on the van der Waals' forces. Greater the molecule mass, greater will be the van der Waals' forces. Hence higher is the boiling point. Thus, the correct order is $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$.